MAT 5173 — Weekly Schedule

Algebra I · Fall 2026 · UT San Antonio

Texts

Primary text

- **Ash** — Ash, *Abstract Algebra: The Basic Graduate Year*, 2002

Secondary reading

- **LiZh** — Li & Zhao, *Introduction to Abstract Algebra*, Les Ulis: EDP Sciences, 2022. ISBN 978-2-7598-2915-6. DOI: 10.1051/978-2-7598-2916-3
- **CLO** — Cox, Little & O'Shea, *Ideals, Varieties, and Algorithms*, 5th ed., Springer, 2025. ISBN 978-3-031-91840-7. DOI: 10.1007/978-3-031-91841-4

Supplementary reading

- **Shafarevich** — Shafarevich, *Basic Algebraic Geometry 1*, 3rd ed., Springer, 2013. ISBN 978-3-642-37955-0. DOI: 10.1007/978-3-642-37956-7
- **Rosebrock** — Rosebrock, *Visual Group Theory*, Springer, 2024. ISBN 978-3-662-69364-3. DOI: 10.1007/978-3-662-69365-0
- **Beardon** — Beardon, *Algebra and Geometry*, Cambridge University Press, 2005. ISBN 978-0-521-81362-4. DOI: 10.1017/CBO9780511800436

Handouts & papers

- **Smith** — Smith, Kahanpaa, Kekalainen & Traves, *An Invitation to Algebraic Geometry*, Springer, 2000. ISBN 978-0-387-98980-8. DOI: 10.1007/978-1-4757-4497-2

Schedule

Date	Topic	Reading
Wed 19 Aug	Lecture — Course launch; groups and subgroups	Ash §1.1; LiZh §1.3-1.4
Mon 24 Aug	Lecture — Permutation groups; cyclic groups; the order of an element	Ash §1.2; LiZh §1.5, §2.1; Rosebrock Ch. 2; E-handout: Complex geometry (passphrase in Canvas)
Wed 26 Aug	Lecture — Cosets, normal subgroups and homomorphisms	Ash §1.3; LiZh §3.1, §3.3
Mon 31 Aug	Lecture — The isomorphism theorems	Ash §1.4; LiZh §3.4
Wed 2 Sep	Lecture — Direct products	Ash §1.5; Beardon Ch. 12
Wed 9 Sep	Lecture — Quotient groups and the isomorphism theorems; direct products	Ash §1.3-1.4; LiZh §3.3, §3.4
Mon 14 Sep	Lecture — Quotient groups and the isomorphism theorems; direct products (continued)	Ash §1.3-1.4; LiZh §3.3, §3.4
Wed 16 Sep	**Q&A** (**Due**: Set 1)	Problem Set 1 (PDF); Activity: Quotients and Homomorphic Images (Colab, ungraded)
Mon 21 Sep	Lecture — Rings: basic definitions and properties; ideals, homomorphisms and quotient rings	Ash §2.1-2.2; LiZh §4.1, §4.5

Date	Topic	Reading
Wed 23 Sep	Lecture — The isomorphism theorems for rings; maximal and prime ideals	Ash §2.3-2.4; LiZh §4.5, §4.7
Mon 28 Sep	Lecture — Polynomial rings; unique factorization	Ash §2.5-2.6; LiZh §5.1
Wed 30 Sep	Lecture — Polynomial rings and unique factorization (continued)	Ash §2.5-2.6; LiZh §5.1
Mon 5 Oct	**Q&A** (**Due**: Set 2)	Problem Set 2 (PDF); Activity: Polynomial Factorization (Colab, ungraded)
Wed 7 Oct	**Exam** — Midterm Exam 1 — Sets 1 and 2	
Wed 14 Oct	Lecture — Principal ideal domains and Euclidean domains; rings of fractions; irreducible polynomials	Ash §2.7-2.9; LiZh §5.2-5.3
Mon 19 Oct	Lecture — Field extensions; degree; simple extensions	Ash §3.1; LiZh §6.1-6.2
Wed 21 Oct	Lecture — Splitting fields; algebraic closures	Ash §3.2-3.3; LiZh §6.3
Mon 26 Oct	Lecture — Separability; normal extensions; finite fields	Ash §3.4-3.5; Ash §6.4; LiZh §6.4
Wed 28 Oct	Lecture — Affine varieties; V(I) and I(V): algebra ↔ geometry	CLO §1.1-1.4; Ash §8.1; Smith Ch. 1 (handout)
Mon 2 Nov	**Q&A** (**Due**: Set 3)	Problem Set 3 (PDF); Activity: Finite Fields and Polynomials (Colab, ungraded)
Wed 4 Nov	Lecture — The coordinate ring and the algebra-geometry dictionary; the Nullstellensatz and the Hilbert basis theorem (both stated, not proved)	CLO §1.4; Ash §8.2-8.3
Mon 9 Nov	Lecture — Plane curves: rational curves; nodes and cusps	Shafarevich §1.1
Wed 11 Nov	Lecture — Fixed fields and Galois groups	Ash §6.1
Mon 16 Nov	Lecture — The fundamental theorem of Galois theory	Ash §6.2; Ash §6.3
Wed 18 Nov	**Q&A** (**Due**: Set 4)	
Mon 23 Nov	Lecture — Survey: the unsolvability of the quintic	Ash §6.8
Mon 30 Nov	*Review* — Review — the exam pool	
Wed 2 Dec	**Exam** — Midterm Exam 2 — Sets 3 and 4	